

Oesophagus

Gastro-oesophageal reflux and hiatus hernia, Barrett's oesophagus and adenocarcinoma, eosinophilic oesophagitis and squamous cell carcinoma (ADH1B, ALDH2, PLCE1)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Barrett's oesophagus and adenocarcinoma have a low-to-typical predisposition: the genomic score for Barrett's or adenocarcinoma is at percentile **14.1**, the adenocarcinoma score at percentile 45.3, and the burden of 58 Barrett's loci at percentile 49.1. For eosinophilic oesophagitis, Sérgio F carries none of the risk alleles of **TSLP** (G/G), **EMSY/LRRC32** (C/C) or SMAD3; the genomic score is at percentile 67.5 and the loci burden at percentile 41.5. For squamous cell carcinoma, **ALDH2*2** is absent (G/G) and so is the PLCE1 risk allele. The reflux genes CRTC1, ATXN2/SH2B3 and SLC39A8 and the hiatus hernia gene WT1 carry no risk alleles.

Unfavourable / mixed. Gastro-oesophageal reflux has discordant scores: the 2024 genomic score is at percentile **83.1**, but the two 2022 scores are at percentiles 6.8 and **10.4**; the consensus is 10.4 and the burden of 98 reflux loci is at percentile 75.0. The **14-variant oesophageal cancer score** is at percentile **84.8**, but the genomic score for the same trait is at percentile 53.7 (consensus 69.2). Some Barrett's loci are in risk homozygosity (**BARX1** A/A, **FOXP1** A/A, CFTR) and **FOXF1** has 1 allele, as in the Cancers One by One report. The protective **ADH1B*2** is absent (C/C), which only matters with alcohol.

What it means. The oesophagus has a typical-to-favourable genetic profile. The point to watch is reflux: the scores disagree and the most recent one points upwards, so persistent heartburn deserves attention (chronic reflux is the route to Barrett's). The oesophageal cancer score at the 85th percentile has few variants and is not confirmed by the genomic score; with zero alcohol and no tobacco, squamous cell carcinoma loses almost all of its known genetic risk.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (Beagle 5.5, public panel of 30x whole genomes) with quality $DR2 \geq 0.8$. Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 20 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 18 from the chip, 2 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 1 SNP needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 11 PGS Catalog scores calculated locally with 5,744,764 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentile ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ($n = 633$) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 246 loci, 207 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Gastro-oesophageal reflux	10.4	3 of 3	44.9	2
Hiatus hernia and other oesophageal diseases	27.5	1 of 2	27.5	1
Barrett's and adenocarcinoma	14.1	1 of 2	14.1	1
Eosinophilic oesophagitis	67.5	1 of 1	67.5	1
Oesophageal cancer (all types)	69.2	2 of 3	53.7	1

In all these scores, a higher percentile means more risk. The hiatus hernia score (46%) and the oesophageal cancer score with 2,001 variants (18%) have low coverage and are left out of the consensus; so is the adenocarcinoma-only score (69%).

Gastro-oesophageal reflux

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Gastro-oesophageal reflux (genomic)	PGS004468 (Jung 2024, 1.06 million variants)	1,025,639 of 1,059,939 (coverage 96%)	83.1	MORE LIKELY	unfavourable
Oesophagitis, reflux and related diseases (genomic)	PGS002063 (Privé 2022, 836,000 variants)	820,027 of 836,413 (coverage 97%)	6.8	LESS LIKELY	favourable
Oesophagitis, reflux and related diseases	PGS001851 (Privé 2022, 14,000 variants)	13,437 of 13,855 (coverage 96%)	10.4	LESS LIKELY	favourable

Hiatus hernia and other oesophageal diseases

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hiatus hernia	PGS000939 (Tanigawa 2022, 2,553 variants)	1,134 of 2,553 (coverage 46%)	74.2	TYPICAL	typical
Other oesophageal diseases (genomic)	PGS004469 (Jung 2024, 1.06 million variants)	1,025,639 of 1,059,939 (coverage 96%)	27.5	TYPICAL	typical

Barrett's and adenocarcinoma

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Barrett's oesophagus or adenocarcinoma (genomic)	PGS003387 (Namba 2022, 602,000 variants)	522,372 of 601,980 (coverage 82%)	14.1	LESS LIKELY	favourable
Oesophageal adenocarcinoma (genomic)	PGS003388 (Namba 2022, 357,000 variants)	275,813 of 356,743 (coverage 69%)	45.3	TYPICAL	typical

Eosinophilic oesophagitis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Eosinophilic oesophagitis (genomic)	PGS012513 (Trimarchi 2026, 1.04 million variants)	1,018,233 of 1,039,900 (coverage 97%)	67.5	TYPICAL	typical

Oesophageal cancer (all types)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Oesophageal cancer (genomic)	PGS000365 (Fritsche 2020, 1.07 million variants)	1,042,126 of 1,070,434 (coverage 97%)	53.7	TYPICAL	typical
Oesophageal cancer	PGS000364 (Fritsche 2020, 2,001 variants)	331 of 2,001 (coverage 18%)	72.5	TYPICAL	typical

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Oesophageal cancer	PGS002298 (Choi 2020, 14 variants)	13 of 14 (coverage 98%)	84.8	MORE LIKELY	unfavourable

4. Key variants by topic

Replicated loci for reflux and hiatus hernia, Barrett's oesophagus and adenocarcinoma, eosinophilic oesophagitis and squamous cell carcinoma (alcohol metabolism, PLCE1, CHEK2). Published alleles with a protective effect were inverted to the risk direction.

Gastro-oesophageal reflux and hiatus hernia

Genes linked to the sphincter, the diaphragm and eating behaviour.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CRTC1	rs10404696		T/T	chip	G (0 copies)	favourable	CRTC1 (neural signalling and appetite): gastro-oesophageal reflux [GWAS Catalog: Acid reflux/GERD, allele G, beta 0.041 [0.03-0.052] unit increase, p=7E-15, PMID 39024449]	Moderate
ATXN2/SH2B3	rs4766578		A/A	chip	T (0 copies)	favourable	ATXN2/SH2B3 region (12q24): gastro-oesophageal reflux [GWAS Catalog: Esophagitis, GERD and related diseases (PheCode 530.1), allele T, beta 0.032 [0.023-0.04] unit increase, p=1E-15, PMID 39024449]	Moderate
SLC39A8	rs13107325	A391T	C/C	chip	T (0 copies)	favourable	A391T of zinc and manganese transporter ZIP8: reflux and several digestive diseases [GWAS Catalog: Gastroesophageal reflux disease, allele C, beta 0.070 unit decrease; risk/increase allele = T, the opposite, p=2E-14, PMID 34187846]	Moderate
WT1	rs11031796		A/A	imputed DR2=0.98	G (0 copies)	favourable	WT1 (diaphragm development): hiatus hernia [GWAS Catalog: Hiatus hernia, allele G, OR 1.07, p=4E-16, PMID 36584111]	Moderate

Barrett's oesophagus and adenocarcinoma

Genes of oesophageal development (FOXF1, BARX1, FOXP1) and bicarbonate secretion.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CDKN2B-AS1	rs1537372		T/T	chip	G (0 copies)	favourable	9p21 (CDKN2B-AS1): Barrett's oesophagus [GWAS Catalog: Barrett's esophagus (PheCode 530.13), allele G, beta 0.131 [0.11-0.16] unit increase, p=3E-26, PMID 39024449]	Moderate
FOXF1	rs9936833		C/T	chip	C (1 copy)	unfavourable (mild)	FOXF1 (16q24, development of the oesophagus): Barrett and adenocarcinoma [GWAS Catalog: Barrett's esophagus, allele C, OR 1.14, p=3E-10, PMID 22961001]	Moderate
SLC22A3	rs622217		T/C	chip	T (1 copy)	unfavourable (mild)	SLC22A3 transporter: Barrett and adenocarcinoma of the oesophagus [GWAS Catalog: Barrett's esophagus or Esophageal adenocarcinoma, allele T, beta 0.130 [0.1-0.16] unit increase, p=3E-18, PMID 35882562]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BARX1	rs11789015	A/A	chip		A (2 copies)	unfavourable	BARX1 (development of oesophagus and stomach): Barrett and adenocarcinoma [GWAS Catalog: Digestive system disease (Barrett's esophagus and esophageal, allele A, OR 1.20, p=1E-9, PMID 24121790]	Moderate
FOXP1	rs2687201	A/A	chip		T (2 copies)	unfavourable	FOXP1: Barrett and adenocarcinoma [GWAS Catalog: Digestive system disease (Barrett's esophagus and esophageal, allele T, OR 1.18, p=5E-9, PMID 24121790]	Moderate
CFTR	rs2188554	A/A	chip		A (2 copies)	unfavourable	CFTR region (bicarbonate secretion): Barrett and adenocarcinoma [GWAS Catalog: Barrett's esophagus or Esophageal adenocarcinoma, allele A, beta 0.128 [0.091-0.165] unit increase, p=3E-11, PMID 35882562]	Weak
NNT	rs72760500	G/G	chip		A (0 copies)	favourable	NNT (mitochondrial transhydrogenase): Barrett and adenocarcinoma [GWAS Catalog: Barrett's esophagus or Esophageal adenocarcinoma, allele A, beta 0.227 [0.17-0.29] unit increase, p=2E-13, PMID 35882562]	Weak

Eosinophilic oesophagitis

Epithelial alarmins (TSLP) and barrier genes, shared with atopy.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TSLP	rs252716	G/G	chip		C (0 copies)	favourable	TSLP/WDR36 (epithelial alarmin): eosinophilic oesophagitis (~1.5x) [GWAS Catalog: Eosinophilic esophagitis, allele C, OR 1.52, p=4E-14, PMID 25407941]	Strong
EMSY/LRRC32	rs61894547	C/C	chip		T (0 copies)	favourable	EMSY/LRRC32 (11q13): eosinophilic oesophagitis and atopy [GWAS Catalog: Eosinophilic esophagitis, allele T, OR 1.92, p=4E-13, PMID 34506852]	Moderate
ANKRD27	rs3815700	T/C	chip		C (1 copy)	unfavourable (mild)	ANKRD27: eosinophilic oesophagitis [GWAS Catalog: Eosinophilic esophagitis, allele C, OR 1.62, p=2E-9, PMID 25407941]	Moderate
KLF13	rs8041227	G/A	chip		G (1 copy)	unfavourable (mild)	KLF13 region: eosinophilic oesophagitis [GWAS Catalog: Eosinophilic esophagitis, allele G, OR 1.52, p=6E-10, PMID 25017104]	Weak
SMAD3	rs56062135	C/C	chip		T (0 copies)	favourable	SMAD3 (TGF- β pathway, oesophageal fibrosis): eosinophilic oesophagitis [GWAS Catalog: Eosinophilic esophagitis, allele T, OR 1.29, p=4E-10, PMID 34506852]	Moderate

Squamous cell carcinoma and alcohol metabolism

Acetaldehyde from alcohol is the main carcinogen for squamous cell carcinoma.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ADH1B	rs1229984	Arg48His	C/C	imputed DR2=0.88	T (0 copies)	neutral	ADH1B*2 (Arg48His): the T allele greatly accelerates the conversion of ethanol to acetaldehyde and protects from heavy consumption; without it, with alcohol, there is more squamous cell carcinoma	Strong
ALDH2	rs671	Glu504Lys	G/G	chip	A (0 copies)	favourable	ALDH2*2 (Glu504Lys): the A allele accumulates acetaldehyde with alcohol and multiplies squamous cell carcinoma of the oesophagus	Strong
PLCE1	rs2274223		A/A	chip	G (0 copies)	favourable	PLCE1: squamous cell carcinoma and cardia cancer [GWAS Catalog: Esophageal cancer, allele G, OR 1.34, p=4E-20, PMID 21642993]	Moderate
CHEK2	rs4822983		C/T	chip	T (1 copy)	unfavourable (mild)	CHEK2 region (22q12): squamous cell carcinoma [GWAS Catalog: Esophageal cancer (squamous cell), allele T, OR 1.27, p=2E-22, PMID 22960999]	Weak

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for reflux, hiatus hernia, Barrett's and adenocarcinoma, eosinophilic oesophagitis and oesophageal cancer. Alleles are oriented towards more risk and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Gastro-oesophageal reflux	98 (92)	88 of 184	91	75.0	typical
Hiatus hernia	8 (7)	7 of 14	7	21.7	typical
Barrett's oesophagus and adenocarcinoma	76 (58)	46 of 116	58	49.1	typical
Eosinophilic oesophagitis	27 (18)	12 of 36	18	41.5	typical
Squamous cell carcinoma and oesophageal cancer	37 (32)	21 of 64	32	45.0	typical

6. Favourable

- **Barrett's and adenocarcinoma:** genomic score at percentile **14.1**; adenocarcinoma at percentile 45.3; loci burden at percentile 49.1; CDKN2B-AS1, SLC22A3 (1 allele) and NNT without relevant risk. **Moderate.**
- **Eosinophilic oesophagitis:** TSLP (G/G), EMSY/LRRC32 (C/C) and SMAD3 without risk alleles; genomic score at percentile 67.5. **Moderate.**
- **ALDH2*2 absent** (G/G) and **PLCE1** without the risk allele (A/A). Strong (ALDH2).
- **Reflux in the classic genes:** CRTC1, ATXN2/SH2B3 and SLC39A8 A391T without risk alleles; hiatus hernia with WT1 without risk and loci at percentile 21.7. **Moderate.**

7. Unfavourable

- **Reflux, discordant scores:** 2024 genomic score at percentile **83.1** versus 6.8 and 10.4 for the 2022 scores; loci burden at percentile 75.0. **Weak to moderate.**
- **Oesophageal cancer, 14-variant score** at percentile **84.8**; not confirmed by the genomic score (53.7). **Weak.**
- **BARX1** A/A, **FOXP1** A/A and **CFTR** A/A in risk homozygosity; **FOXF1** C/T with 1 allele. Effects of about 1.1–1.2× each. **Moderate.**
- **ADH1B*2 absent** (C/C): no protection against heavy drinking; it only matters with alcohol. Moderate (conditional).
- **ANKRD27** and **KLF13** (eosinophilic oesophagitis) and **CHEK2** (squamous cell carcinoma) with 1 allele. **Weak.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
Chronic Inflammation	Reflux with a mixed signal (consensus at percentile 10.4); eosinophilic oesophagitis without the main risk alleles	Typical
Somatic Mutation Accumulation	Barrett's and adenocarcinoma with the genomic score at percentile 14.1; Barrett's is a lesion of reflux-induced somatic mutations	Favourable
DNA Damage and DNA Repair Deficit	Acetaldehyde from alcohol: normal ALDH2, slow ADH1B; risk only with alcohol	Conditional
Altered Intercellular Communication	Epithelial alarmins (TSLP) of eosinophilic oesophagitis without risk alleles	Favourable
Extracellular Matrix Remodelling	SMAD3 (oesophageal fibrosis in eosinophilic oesophagitis) without the risk allele	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Heartburn or regurgitation more than twice a week** for months justifies a consultation: chronic reflux is the route to Barrett's, and the most recent genomic reflux score is high. *Suggestion (not applied)*.
2. **Simple measures for reflux**: dinner 3 hours before bed, smaller evening meals, and raising the head of the bed if there are night-time symptoms. *Suggestion (not applied)*.
3. **Endoscopy** only if there are persistent symptoms, difficulty swallowing or food getting stuck (this is how eosinophilic oesophagitis shows itself). Without symptoms, genetics does not justify screening. *Suggestion (not applied)*.
4. **Proton pump inhibitors**, if needed: CYP2C19 *1/*1 gives normal efficacy; use the lowest effective dose for the time needed. *Suggestion (not applied)*.
5. **Already in the plan and protective**: zero alcohol, no tobacco, high fibre and glucose control. *Suggestion (not applied)*.

10. Limitations

- The reflux scores disagree (from 6.8 to 83.1); the consensus is low, but the most recent score points upwards.
- The oesophageal cancer score at the 85th percentile has only 14 variants; the genomic score for the same trait is typical.
- Adenocarcinoma and squamous cell carcinoma are different diseases; the general oesophageal cancer score mixes them.
- Reflux and the oesophagus depend heavily on the environment (meals, alcohol, tobacco, sleep, medicines).
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local imputation (Beagle 5.5, public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; for palindromic SNPs the + strand is assumed (the GWAS Catalog and PGS Catalog convention). Published alleles with a protective effect were inverted to the risk direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (11 scores: reflux, hiatus hernia, Barrett's, adenocarcinoma, eosinophilic oesophagitis and oesophageal cancer). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 5 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Ong 2022 (reflux), Gharahkhani 2016 and Ong 2022 (Barrett's and adenocarcinoma), Kottyan 2014 and 2021 (eosinophilic oesophagitis), Wu 2011 and Abnet 2010 (squamous cell carcinoma and PLCE1), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Gastro-oesophageal reflux and hiatus hernia	CRTC1	rs10404696	T/T	chip	G	0	+ strand	without the risk allele
Gastro-oesophageal reflux and hiatus hernia	ATXN2/SH2B3	rs4766578	A/A	chip	T	0	+ strand (palindromic; + assumed)	without the risk allele
Gastro-oesophageal reflux and hiatus hernia	SLC39A8	rs13107325	C/C	chip	T	0	+ strand	without the risk allele
Gastro-oesophageal reflux and hiatus hernia	WT1	rs11031796	A/A	imputed DR2=0.98	G	0	+ strand	without the risk allele
Barrett's oesophagus and adenocarcinoma	CDKN2B-AS1	rs1537372	T/T	chip	G	0	+ strand	without the risk allele
Barrett's oesophagus and adenocarcinoma	FOXF1	rs9936833	C/T	chip	C	1	+ strand	1 copy of the risk allele
Barrett's oesophagus and adenocarcinoma	SLC22A3	rs622217	T/C	chip	T	1	+ strand	1 copy of the risk allele
Barrett's oesophagus and adenocarcinoma	BARX1	rs11789015	A/A	chip	A	2	+ strand	2 copies of the risk allele
Barrett's oesophagus and adenocarcinoma	FOXP1	rs2687201	A/A	chip	T	2	complement (– strand)	2 copies of the risk allele
Barrett's oesophagus and adenocarcinoma	CFTR	rs2188554	A/A	chip	A	2	+ strand	2 copies of the risk allele
Barrett's oesophagus and adenocarcinoma	NNT	rs72760500	G/G	chip	A	0	+ strand	without the risk allele
Eosinophilic oesophagitis	TSLP	rs252716	G/G	chip	C	0	+ strand (palindromic; + assumed)	without the risk allele
Eosinophilic oesophagitis	EMSY/LRRC32	rs61894547	C/C	chip	T	0	+ strand	without the risk allele
Eosinophilic oesophagitis	ANKRD27	rs3815700	T/C	chip	C	1	+ strand	1 copy of the risk allele
Eosinophilic oesophagitis	KLF13	rs8041227	G/A	chip	G	1	+ strand	1 copy of the risk allele
Eosinophilic oesophagitis	SMAD3	rs56062135	C/C	chip	T	0	+ strand	without the risk allele
Squamous cell carcinoma and alcohol metabolism	ADH1B	rs1229984	C/C	imputed DR2=0.88	T	0	+ strand	without the protective allele (common genotype)
Squamous cell carcinoma and alcohol metabolism	ALDH2	rs671	G/G	chip	A	0	+ strand	without the risk allele
Squamous cell carcinoma and alcohol metabolism	PLCE1	rs2274223	A/A	chip	G	0	+ strand	without the risk allele
Squamous cell carcinoma and alcohol metabolism	CHEK2	rs4822983	C/T	chip	T	1	+ strand	1 copy of the risk allele